

Math 242, S20
Final Exam

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 9 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	12	
6	12	
7	12	
8	12	
9	12	
Total	100	

Good Luck !

Problem 1. (10 points) Find the particular solution of the initial value problem:

$$(1+x) \frac{dy}{dx} + 3y = \frac{x \cos x}{(1+x)^2}, \quad y(0) = 3$$

Solution

$$\frac{dy}{dx} + \underbrace{\left(\frac{3}{1+x}\right)}_{P(x)} y = \underbrace{\left(\frac{x \cos x}{(1+x)^3}\right)}_{Q(x)}$$

Integrating factor:

$$\begin{aligned} P(x) &= e^{\int P(x) dx} = e^{\int \frac{3}{1+x} dx} = e^{3 \ln(1+x)} \\ &= (1+x)^3 \end{aligned}$$

Thus

$$\begin{aligned} \frac{d}{dx} [P(x) y(x)] &= P(x) Q(x) \\ (1+x)^3 y(x) &= \int (1+x)^3 \cdot \frac{x \cos x}{(1+x)^3} dx \\ &= \int x \cos x dx \\ &= x \sin x + \cos x + C \end{aligned}$$

$$\Rightarrow y(x) = \frac{x \sin x + \cos x + C}{(1+x)^3}$$

$$y(0) = 3 \Rightarrow \frac{1+C}{1^3} = 3 \Rightarrow C = 2$$

$$\Rightarrow y(x) = \frac{x \sin x + \cos x + 2}{(1+x)^3}$$

Problem 2. (10 points) Given a first-order differential equation

$$(2 + ye^{xy})dx + (2y + xe^{xy})dy = 0.$$

(a) Verify that the above equation is **exact**.

$$M(x, y) = 2 + ye^{xy}, \quad N(x, y) = 2y + xe^{xy}$$

$$\left. \begin{aligned} \frac{\partial M}{\partial y} &= e^{xy} + xy e^{xy} \\ \frac{\partial N}{\partial x} &= e^{xy} + xy e^{xy} \end{aligned} \right\} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{exact.}$$

(b) Find a general solution in the form of $F(x, y) = C$ for the above equation.

$$\begin{aligned} F(x, y) &= \int M(x, y) dx \\ &= \int (2 + ye^{xy}) dx \\ &= 2x + e^{xy} + g(y) \end{aligned}$$

$$\begin{aligned} \frac{\partial F}{\partial y} &= N(x, y) \Rightarrow \frac{\partial}{\partial y} [2x + e^{xy} + g(y)] = 2y + xe^{xy} \\ xe^{xy} + g'(y) &= 2y + xe^{xy} \\ \Rightarrow g'(y) &= 2y \Rightarrow g(y) = \int 2y dy = y^2 + C_1 \end{aligned}$$

Thus

$$F(x, y) = 2x + e^{xy} + y^2 = C$$

Problem 3. (10 points) Find a general solution for the following differential equations:

$$\frac{dy}{dx} = (3x + 4y)^2.$$

$$\text{let } V = 3x + 4y \Rightarrow y = \frac{1}{4}V - \frac{3}{4}x$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{4} \frac{dV}{dx} - \frac{3}{4}$$

Thus

$$\frac{1}{4} \frac{dV}{dx} - \frac{3}{4} = V^2$$

$$\frac{dV}{dx} = 3 + 4V^2$$

$$\frac{dV}{1 + \frac{4}{3}V^2} = 3dx$$

$$\int \frac{1}{1 + (\frac{2}{\sqrt{3}}V)^2} dV = \int 3dx$$

$$\frac{\sqrt{3}}{2} \arctan\left(\frac{2}{\sqrt{3}}V\right) = 3x + C_1$$

$$\arctan\left(\frac{2}{\sqrt{3}}V\right) = 2\sqrt{3}x + C$$

$$V = \frac{\sqrt{3}}{2} \tan(2\sqrt{3}x + C)$$

$$\Rightarrow 3x + 4y = \frac{\sqrt{3}}{2} \tan(2\sqrt{3}x + C)$$

$$\Rightarrow y(x) = \frac{\sqrt{3}}{8} \tan(2\sqrt{3}x + C) - \frac{3}{4}x$$

Problem 4. (10 points) A pitcher of buttermilk initially at 30°C is to be cooled by setting it on the front porch, where the temperature is 0°C . The temperature of the buttermilk has dropped to 20°C after 10 minutes.

(a) Write down the differential equation for the temperature of the buttermilk $T(t)$ according to the Newton's law of cooling, then find its general solution and determine all constants in the solution using the given conditions.

$$\frac{dT}{dt} = -k(T - A) \quad \begin{matrix} \text{environmental temperature} \\ A = 0 \end{matrix}$$

$$= -kT$$

$$\Rightarrow T(t) = C e^{-kt}$$

$$T(0) = 30 \Rightarrow C = 30 \Rightarrow T(t) = 30 e^{-kt}$$

$$T(10) = 20 \Rightarrow 20 = 30 e^{-k(10)}$$

$$\Rightarrow e^{-10k} = \frac{20}{30}$$

$$-10k = \ln \frac{2}{3} \Rightarrow k = -\frac{1}{10} \ln \frac{2}{3} = \frac{1}{10} \ln \frac{3}{2} \approx 0.04055$$

$$\Rightarrow T(t) = 30 e^{-(\frac{1}{10} \ln \frac{3}{2})t}$$

(b) How long does it take for the temperature of the buttermilk to drop to 5°C ?

$$T(t) = 5$$

$$\Rightarrow 30 e^{-(\frac{1}{10} \ln \frac{3}{2})t} = 5$$

$$-(\frac{1}{10} \ln \frac{3}{2})t = \ln \frac{5}{30}$$

$$t = \frac{10 \ln 6}{\ln(\frac{3}{2})} \approx 44.19 \text{ mins}$$

Problem 5. (12 points)

(a) Given an initial value problem

$$\frac{dy}{dx} = -y + x + 1, \quad y(0) = 2.$$

Apply **Euler's method** to approximate the solution of the above problem on the interval $[0, 0.3]$ with step size $h = 0.1$.

$$x_0 = 0, h = 0.1, y_0 = 2, f(x, y) = -y + x + 1.$$

$$n=1: x_1 = x_0 + h = 0 + 0.1 = 0.1$$

$$\begin{aligned} y_1 &= y_0 + h f(x_0, y_0) = y_0 + h(-y_0 + x_0 + 1) \\ &= 2 + 0.1(-2 + 0 + 1) = 1.9. \end{aligned}$$

$$n=2: x_2 = x_1 + h = 0.2$$

$$y_2 = y_1 + h f(x_1, y_1) = 1.9 + 0.1(-1.9 + 0.1 + 1) = 1.82$$

$$n=3: x_3 = x_2 + h = 0.3$$

$$\begin{aligned} y_3 &= y_2 + h f(x_2, y_2) = 1.82 + 0.1(-1.82 + 0.1 + 1) \\ &= 1.758 \end{aligned}$$

(b) Show that the three functions x, x^2, e^x are linearly independent in any open interval by computing the Wronskian $W(x, x^2, e^x)$.

$$\begin{aligned} W(x, x^2, e^x) &= \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix} \\ &= x \begin{vmatrix} 2x & e^x \\ 2 & e^x \end{vmatrix} - 1 \cdot \begin{vmatrix} x^2 & e^x \\ 2 & e^x \end{vmatrix} \\ &= x(2xe^x - 2e^x) - (x^2e^x - 2e^x) \\ &= x^2e^x - 2xe^x + 2e^x \\ &= (x^2 - 2x + 2)e^x \\ &= [(x+1)^2 + 1]e^x \geq e^x > 0 \end{aligned}$$

Problem 6. (12 points) Given a nonhomogeneous linear third-order differential equation:

$$y^{(3)} + y' = -2\sin x.$$

(a) Find the complementary solution y_c of the above nonhomogeneous equation (i.e., the general solution of the associated homogeneous equation).

$$\begin{aligned} y_c^{(3)} + y_c' &= 0 \\ \text{characteristic equation: } r^3 + r &= 0 \\ r(r^2 + 1) &= 0 \\ \Rightarrow r_1 = 0, r_2 = i, r_3 = -i \\ \Rightarrow y_c(x) &= C_1 + C_2 \cos x + C_3 \sin x \end{aligned}$$

(b) Find a particular solution y_p of the nonhomogeneous equation.

$$\begin{aligned} y_p(x) &= x^5 (A \cos x + B \sin x) \\ &\stackrel{!}{=} Ax \cos x + Bx \sin x \\ y_p'(x) &= A \cos x + B \sin x - Ax \sin x + Bx \cos x \\ y_p''(x) &= -2A \sin x + 2B \cos x - Ax \cos x - Bx \sin x \\ y_p^{(3)}(x) &= -3A \cos x - 3B \sin x + Ax \sin x - Bx \cos x \\ y_p^{(3)}(x) + y_p'(x) &= 2A \cos x - 2B \sin x \stackrel{!}{=} -2 \sin x \\ \Rightarrow \begin{cases} -2A = 0 \\ -2B = -2 \end{cases} &\Rightarrow \begin{cases} A = 0 \\ B = 1 \end{cases} \\ \Rightarrow y_p(x) &= x \sin x. \end{aligned}$$

Problem 7. (12 points)

(a) Find $\mathcal{L}^{-1}\left\{\ln \frac{s+3}{s-3}\right\}$ using the differentiation of transform, $f(t) = -\frac{1}{t}\mathcal{L}^{-1}\{F'(s)\}$.

$$F(s) = \ln \frac{s+3}{s-3} = \ln(s+3) - \ln(s-3)$$

$$\begin{aligned}\mathcal{L}^{-1}\left\{\ln \frac{s+3}{s-3}\right\} &= -\frac{1}{t}\mathcal{L}^{-1}\left\{\frac{d}{ds}[\ln(s+3) - \ln(s-3)]\right\} \\ &= -\frac{1}{t}\mathcal{L}^{-1}\left\{\frac{1}{s+3} - \frac{1}{s-3}\right\} \\ &= -\frac{1}{t}(e^{-3t} - e^{3t}) \\ &= \frac{1}{t}(e^{3t} - e^{-3t})\end{aligned}$$

(b) Find $\mathcal{L}^{-1}\left\{\frac{6}{s^2(s^2+9)}\right\}$ using the convolution property, $\mathcal{L}^{-1}\{F(s)G(s)\} = f(t) * g(t)$.

$$F(s) = \frac{2}{s^2}, \quad G(s) = \frac{3}{s^2+9}$$

$$\Rightarrow f(t) = 2t, \quad g(t) = \sin 3t$$

$$\begin{aligned}\Rightarrow \mathcal{L}^{-1}\left\{\frac{6}{s^2(s^2+9)}\right\} &= (2t) * (\sin 3t) \rightarrow \left(-\frac{1}{3}\cos 3t\right)' \\ &= \int_0^t 2(t-\tau) \sin 3\tau d\tau \\ &= -\frac{2}{3}(t-\tau) \cos 3\tau \Big|_{\tau=0}^{\tau=t} \\ &\quad - \int_0^t \frac{2}{3} \cos 3\tau d\tau \\ &= \frac{2}{3}t - \left[\frac{2}{9}\sin 3\tau\right]_{\tau=0}^{\tau=t} \\ &= \frac{2}{3}t - \frac{2}{9}\sin 3t\end{aligned}$$

Problem 8. (12 points) Use Laplace transform to solve the following initial value problem:

$$x'' + 2x' + 5x = 0; \quad x(0) = 3, \quad x'(0) = -1$$

$$\text{Let } X(s) = \mathcal{L}\{x(t)\}$$

$$\mathcal{L}\{x'\} = sX(s) - x(0) = sX(s) - 3$$

$$\mathcal{L}\{x''\} = s^2X(s) - sx(0) - x'(0) = s^2X(s) - 3s + 1$$

thus

$$\mathcal{L}\{x'' + 2x' + 5x\} = \mathcal{L}\{0\}$$

$$\mathcal{L}\{x''\} + 2\mathcal{L}\{x'\} + 5\mathcal{L}\{x\} = 0$$

$$[s^2X(s) - 3s + 1] + 2[sX(s) - 3] + 5X(s) = 0$$

$$(s^2 + 2s + 5)X(s) = 3s + 5$$

$$X(s) = \frac{3s + 5}{s^2 + 2s + 5}$$

$$\Rightarrow x(t) = \mathcal{L}^{-1}\left\{\frac{3s + 5}{s^2 + 2s + 5}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{3s + 5}{(s+1)^2 + 4}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{2(s+1)}{(s+1)^2 + 4}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{(s+1)^2 + 4}\right\}$$

$$= 3e^{-t}\cos 2t + e^{-t}\sin 2t$$

Problem 9. (12 points)

(a) Transform the system into an equivalent system of four first-order differential equations:

$$x'' - 3x' + 4x - 2y = e^t, \quad 2y'' + 2y' - 3x - y = \sin 2t - 1.$$

let $x = x_1, x' = x_2, y = x_3, y' = x_4.$

$$\begin{cases} x_1' = x_2 \\ x_2' = 3x_2 - 4x_1 + 2x_3 + e^t \\ x_3' = x_4 \\ x_4' = -x_4 + \frac{3}{2}x_1 + \frac{1}{2}x_3 + \frac{1}{2}(\sin 2t - 1) \end{cases}$$

(b) Solve the following system of equations:

$$\begin{matrix} x' = -3x - 4y, & y' = 2x + y \\ \textcircled{1} & \textcircled{2} \end{matrix}$$

From $\textcircled{1}$, $-4y = x' + 3x$
 $y = -\frac{1}{4}x' - \frac{3}{4}x \Rightarrow y' = -\frac{1}{4}x'' - \frac{3}{4}x'$

put into $\textcircled{2}$,

$$-\frac{1}{4}x'' - \frac{3}{4}x' = 2x + (-\frac{1}{4}x' - \frac{3}{4}x)$$

$$-\frac{1}{4}x'' - \frac{1}{2}x' - \frac{5}{4}x = 0$$

$$x'' + 2x' + 5 = 0 \quad (r^2 + 2r + 5 = 0)$$

$$\Rightarrow x(t) = e^{-t}(C_1 \cos 2t + C_2 \sin 2t) \Rightarrow r = -1 \pm 2i$$

$$x'(t) = e^{-t} [(-C_1 + 2C_2) \cos 2t + (-2C_1 - C_2) \sin 2t]$$

thus $y(t) = -\frac{1}{4}x' - \frac{3}{4}x$

$$= e^{-t} \left[(-\frac{1}{2}C_1 - \frac{1}{2}C_2) \cos 2t + (\frac{1}{2}C_1 - \frac{1}{2}C_2) \sin 2t \right]$$